IBM APPLIED DATA SCIENCE CAPSTONE

Data Science Capstone Project Report

Predicting whether the SpaceX Falcon 9 first stage will land

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GitHub: https://github.com/khoalee252006/spacex-falcon9-capstone

Project GitHub Repository

Repository: https://github.com/khoalee252006/spacex-falcon9-capstone

1	**Data Collection – SpaceX API**	https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/01_data_collection_api.ipynb
2	**Data Collection – Web Scraping**	https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/02_data_collection_webscraping.ipynb
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GitHub: https://github.com/khoalee252006/spacex-falcon9-capstone

Executive Summary

Methods

- Data collection: SpaceX REST API + web scraping (Wikipedia)
- Data wrangling: missing values, landing label Class (1 / 0), one-hot encoding
- EDA: pandas, matplotlib / seaborn charts and SQL (SQLite)
- Interactive analytics: Folium map and Plotly Dash dashboard
- Machine learning: Logistic Regression, SVM, Decision Tree, KNN with GridSearchCV (cv = 10)

Key results

- 90 Falcon 9 launches (2010–2020); 60 landed = 66.7% success
- Success grew from 0% (2010–2013) to 90% in 2019
- Best common orbit: VLEO 86%; weakest: GTO 52%
- When a landing was attempted, 87.0% succeeded
- Logistic Regression, SVM, KNN: 83.3% test accuracy (tie); SVM has the best validation accuracy (84.8%)

GitHub: https://github.com/khoalee252006/spacex-falcon9-capstone

Introduction

Background

- SpaceX advertises Falcon 9 launches at about $62 million.
- Other providers charge $165 million or more per launch.
- Most of the saving comes from landing and reusing the first stage.
- A new company ("Space Y") wants to bid against SpaceX.

Problem statement

- Can we predict if the Falcon 9 first stage will land successfully?
- If yes, we can estimate the real cost of a launch.

Questions we answer

- How do payload mass, orbit, launch site and flight number affect landing?
- Did landing success improve over time?
- Where are the launch sites, and why there?
- Which machine-learning model predicts landing best?

GitHub: https://github.com/khoalee252006/spacex-falcon9-capstone

METHODOLOGY

Data Collection – SpaceX API



Key phrases

- requests.get() → JSON response
- pd.json_normalize() to flatten the JSON
- Helper look-ups: BoosterVersion, LaunchSite, PayloadMass, Orbit, Outcome, Block, Serial…
- Missing PayloadMass (5) → filled with mean (6,123.55 kg)

Note on data source

- The live API returned HTTP 522/525 errors (server down) in Oct 2026.
- The code tries the live API first, then falls back to saved copies of the same API responses: IBM's static launch JSON + archived responses (web.archive.org).
- Check: 15 of 17 columns are identical to IBM's reference dataset_part_1.csv.

GitHub: https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/01_data_collection_api.ipynb

METHODOLOGY

Data Collection – Web Scraping



Key phrases

- Page: "List of Falcon 9 and Falcon Heavy launches" – fixed old revision used by the IBM lab (June 2021), so results repeat
- 9 launch tables found; flight number is in the <th> cell, other values in <td> cells
- Columns: Flight No., Date, Time, Version Booster, Launch site, Payload, Payload mass, Orbit, Customer, Launch outcome, Booster landing
- Cleaning: remove footnote marks like [12], parse "4,700 kg" → 4700, parse dates
- Result: 121 launches from 04 Jun 2010 to 06 Jun 2021

GitHub: https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/02_data_collection_webscraping.ipynb

METHODOLOGY

Data Wrangling



- PayloadMass: 5 missing → replaced by mean
- LandingPad: 26 missing (28.9%) → kept, means "no landing pad used"
- Good outcomes (Class = 1): True ASDS (drone ship), True RTLS (ground pad), True Ocean
- Bad outcomes (Class = 0): False ASDS / RTLS / Ocean, None ASDS, None None
- **Success rate = mean(Class) = 66.7%**
- Output: dataset_part_2.csv (+ Class) and dataset_part_3.csv (80 numeric features)

Outcome	Launches	Class
True ASDS	41	1
None None	19	0
True RTLS	14	1
False ASDS	6	0
True Ocean	5	1
False Ocean	2	0
None ASDS	2	0
False RTLS	1	0

GitHub: https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/03_data_wrangling.ipynb

METHODOLOGY

EDA with Data Visualization

Chart type	Chart	Why we use it
Scatter	Flight Number vs. Payload Mass	Does experience or payload change landing?
Scatter	Flight Number vs. Launch Site	Do sites improve over time?
Scatter	Payload Mass vs. Launch Site	Which site handles heavy payloads?
Bar	Success Rate by Orbit	Which orbit is easiest to land from?
Scatter	Flight Number vs. Orbit	Does each orbit improve with flights?
Scatter	Payload Mass vs. Orbit	Is heavy payload a problem in some orbits?
Line	Yearly Success Trend	Is SpaceX getting better each year?

Points are colored by Class: green = landed, red = not landed. Tools: pandas, matplotlib, seaborn.

GitHub: https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/04_eda_visualization.ipynb

METHODOLOGY

EDA with SQL

#	Question	Key SQL
1	Unique launch sites	SELECT DISTINCT
2	5 records with site starting 'CCA'	LIKE 'CCA%' LIMIT 5
3	Total payload for NASA (CRS)	SUM()
4	Average payload of F9 v1.1	AVG()
5	First successful ground-pad landing	MIN(Date)
6	Drone-ship success, payload 4000–6000 kg	AND, > <
7	Count of mission successes / failures	COUNT(*) GROUP BY
8	Boosters with maximum payload	subquery MAX()
9	2015 drone-ship failures by month	substr(Date)
10	Rank landing outcomes 2010-06-04 → 2017-03-20	BETWEEN, ORDER BY DESC

Data: IBM course table Spacex.csv (101 Falcon 9 launches) loaded into SQLite with pandas.to_sql().

GitHub: https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/05_eda_sql.ipynb

METHODOLOGY

Interactive Visual Analytics – Folium & Plotly Dash

Folium map

- folium.Circle + Marker: each launch site
- MarkerCluster: green = landed, red = not landed
- MousePosition: read coordinates on the map
- PolyLine + distance label: site → coast, railway, highway, city
- Nearest features found with OpenStreetMap (Overpass API) + haversine distance

GitHub: https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/06_interactive_folium_map.ipynb

Plotly Dash dashboard

- Dropdown: All sites or one launch site
- Pie chart: successful launches by site / success vs. failure
- RangeSlider: choose a payload range (kg)
- Scatter: payload vs. launch outcome, colored by booster version
- Callbacks update the charts when the user changes inputs

GitHub: https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/07_dash_app.py

Predictive Analysis – Methodology

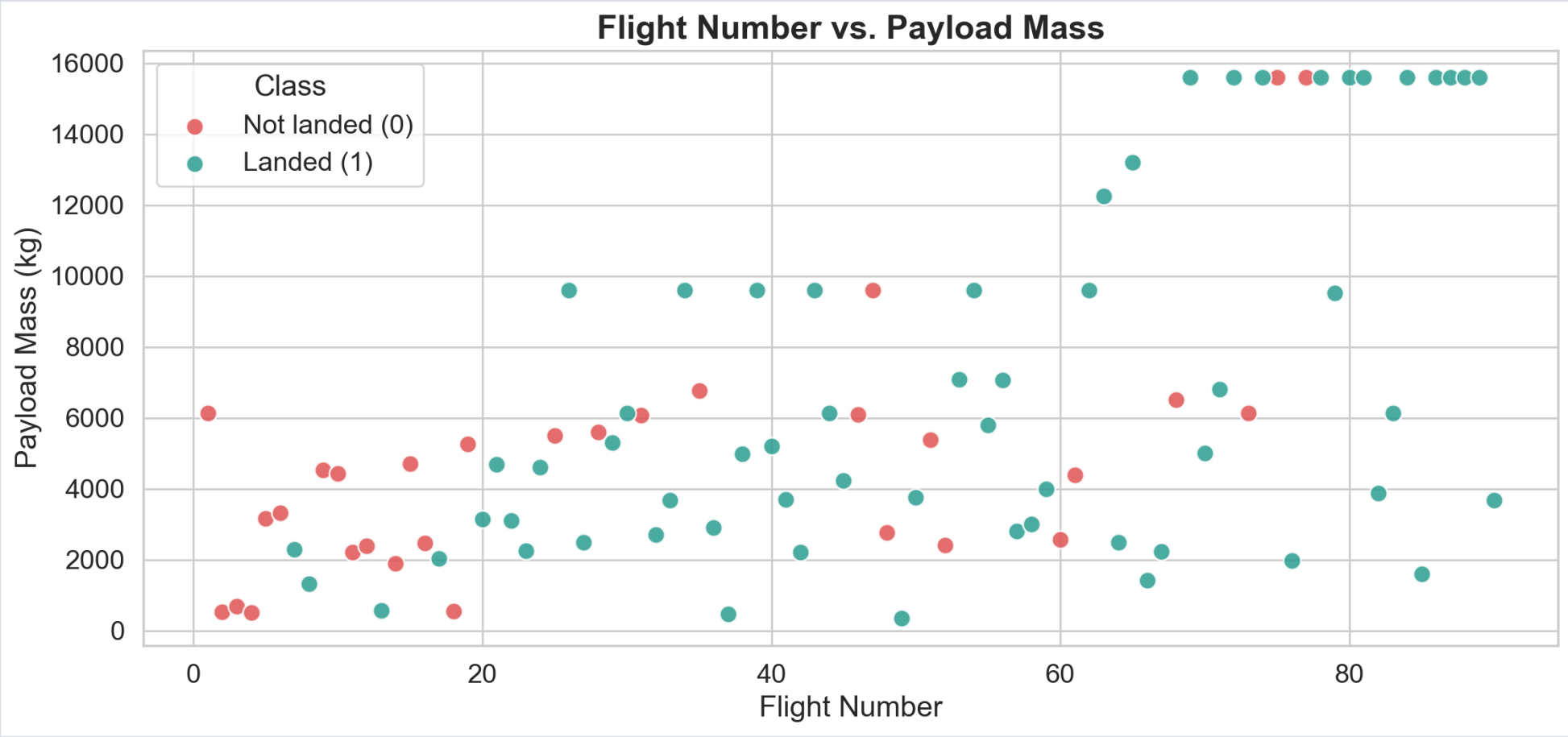


Model	Parameters searched
Logistic Regression	C = 0.01, 0.1, 1
SVM	kernel = linear / rbf / poly / sigmoid; C, gamma = logspace(-3, 3, 5)
Decision Tree	criterion, splitter, max_depth 2–18, max_features, min_samples_leaf / split
KNN	n_neighbors 1–10, algorithm, p = 1 / 2

Extra check: the test set has only 18 launches, so we also run 10-fold stratified cross-validation on all 90 launches.

GitHub: https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/08_predictive_analysis.ipynb

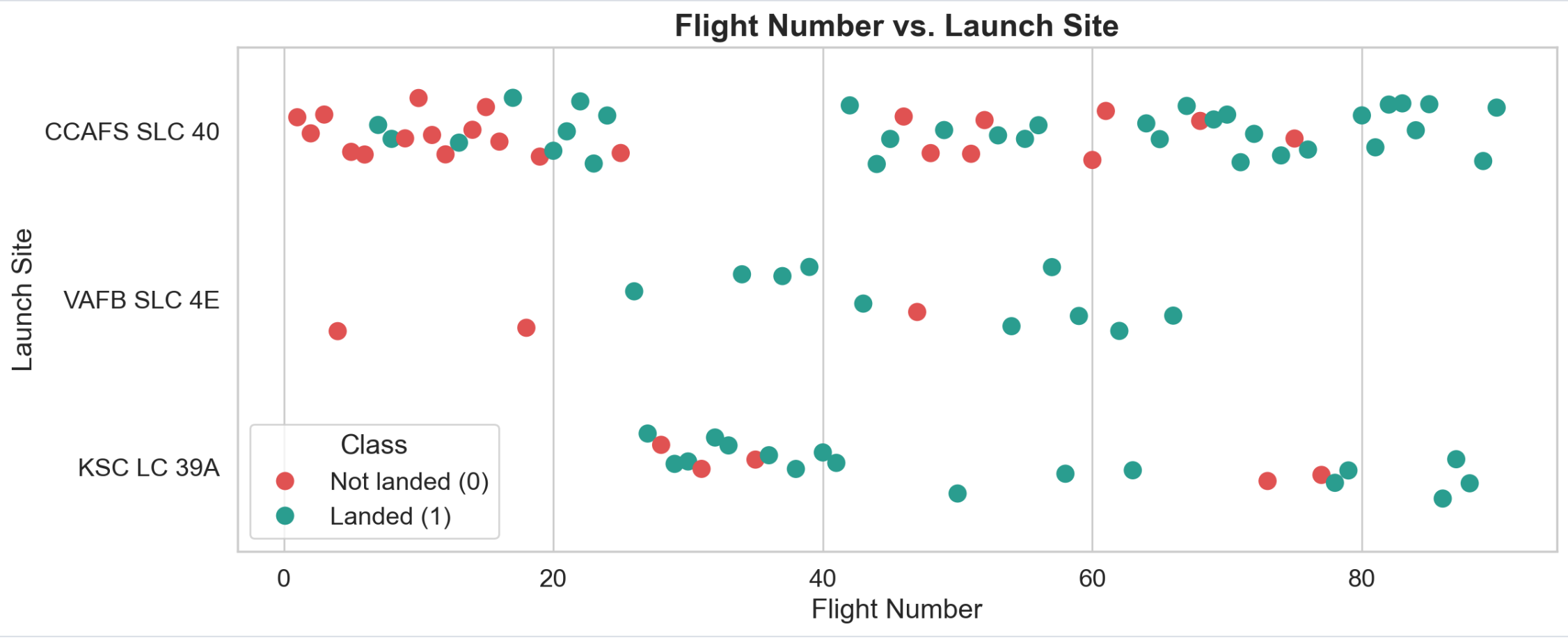
Flight Number vs. Payload Mass



Insight: Later flights land more often (43% for flights 1–30 vs. 83% for flights 61–90). Heavy payloads (> 10,000 kg) still landed 13 of 15 times.

GitHub: https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/04_eda_visualization.ipynb

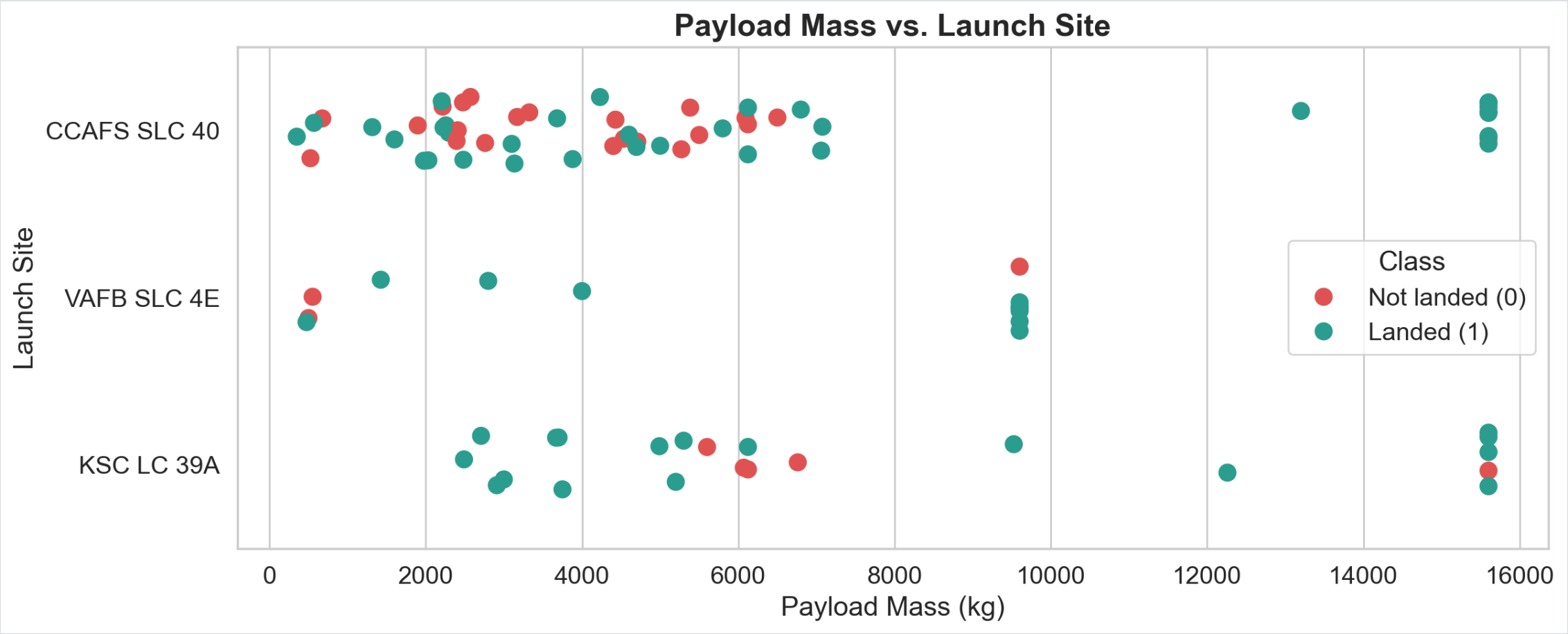
Flight Number vs. Launch Site



Insight: CCAFS SLC 40 has the most launches (55) and the lowest rate (60%): its first 20 launches had 35% success, later ones 74%.

GitHub: https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/04_eda_visualization.ipynb

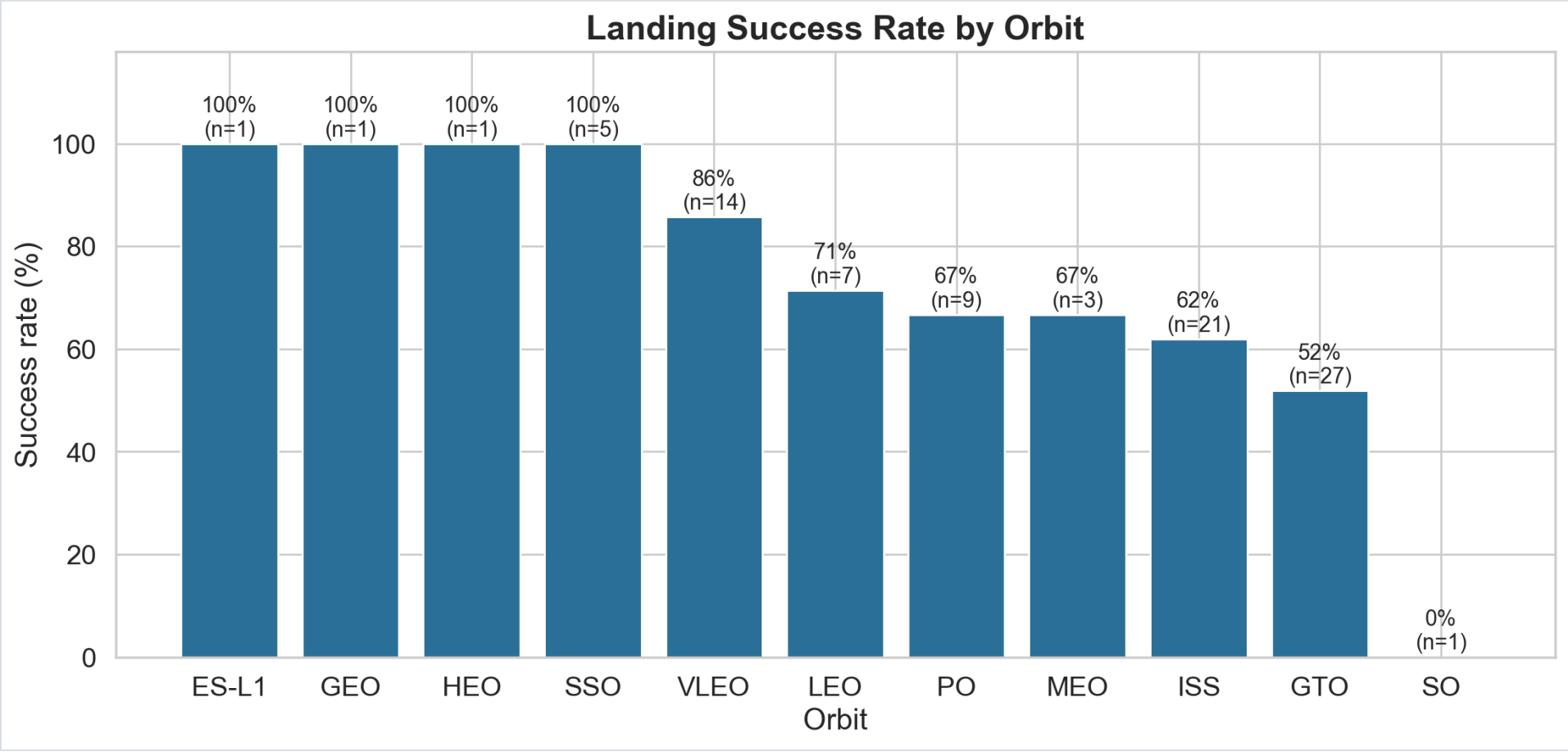
Payload Mass vs. Launch Site



Insight: VAFB SLC 4E never launched more than 9,600 kg. All payloads above 10,000 kg flew from Florida, with high success.

GitHub: https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/04_eda_visualization.ipynb

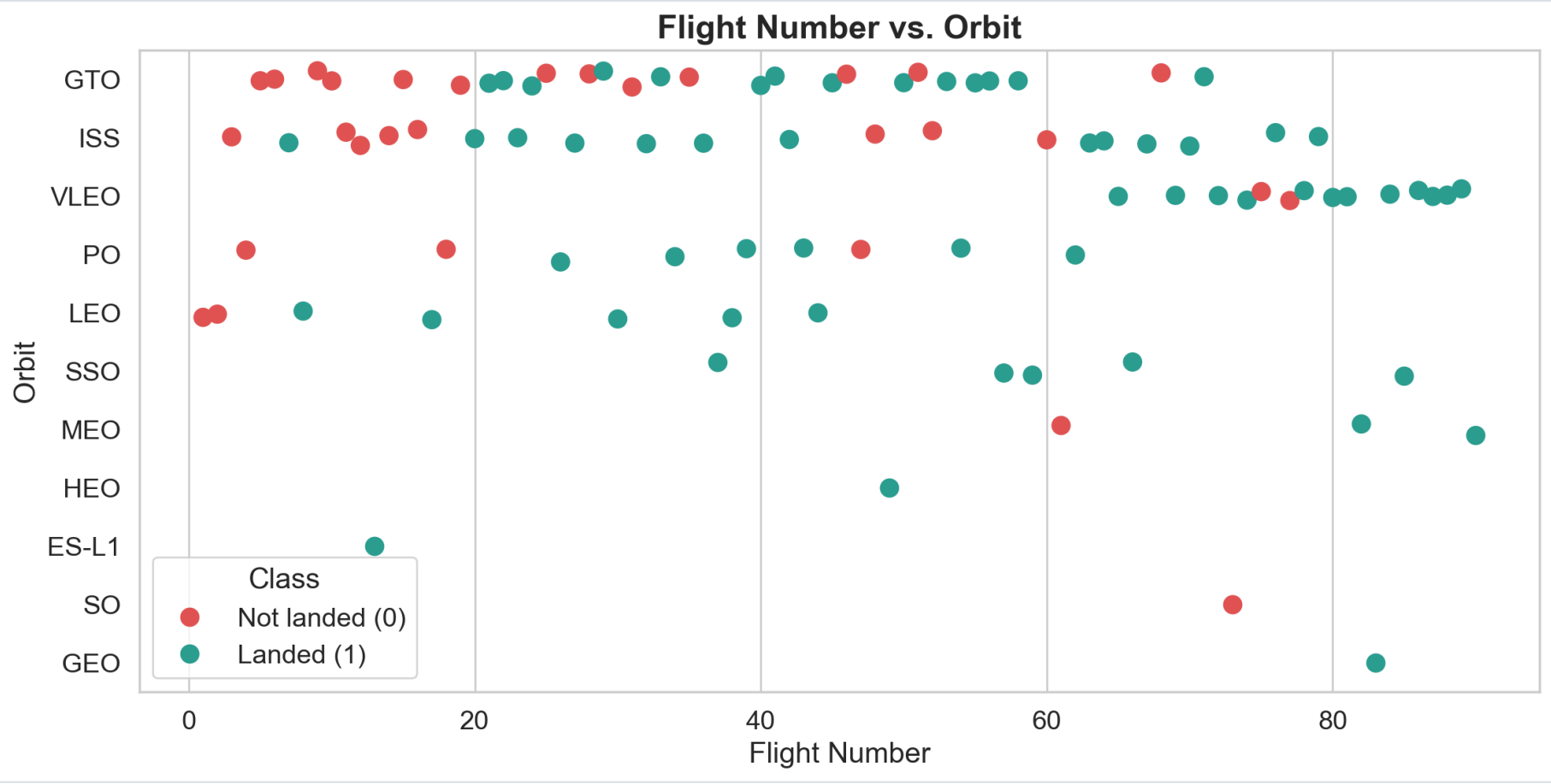
Success Rate by Orbit Type



Insight: SSO, ES-L1, GEO and HEO show 100% but have very few launches. Among common orbits, VLEO is best (86%, n=14) and GTO is weakest (52%, n=27).

GitHub: https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/04_eda_visualization.ipynb

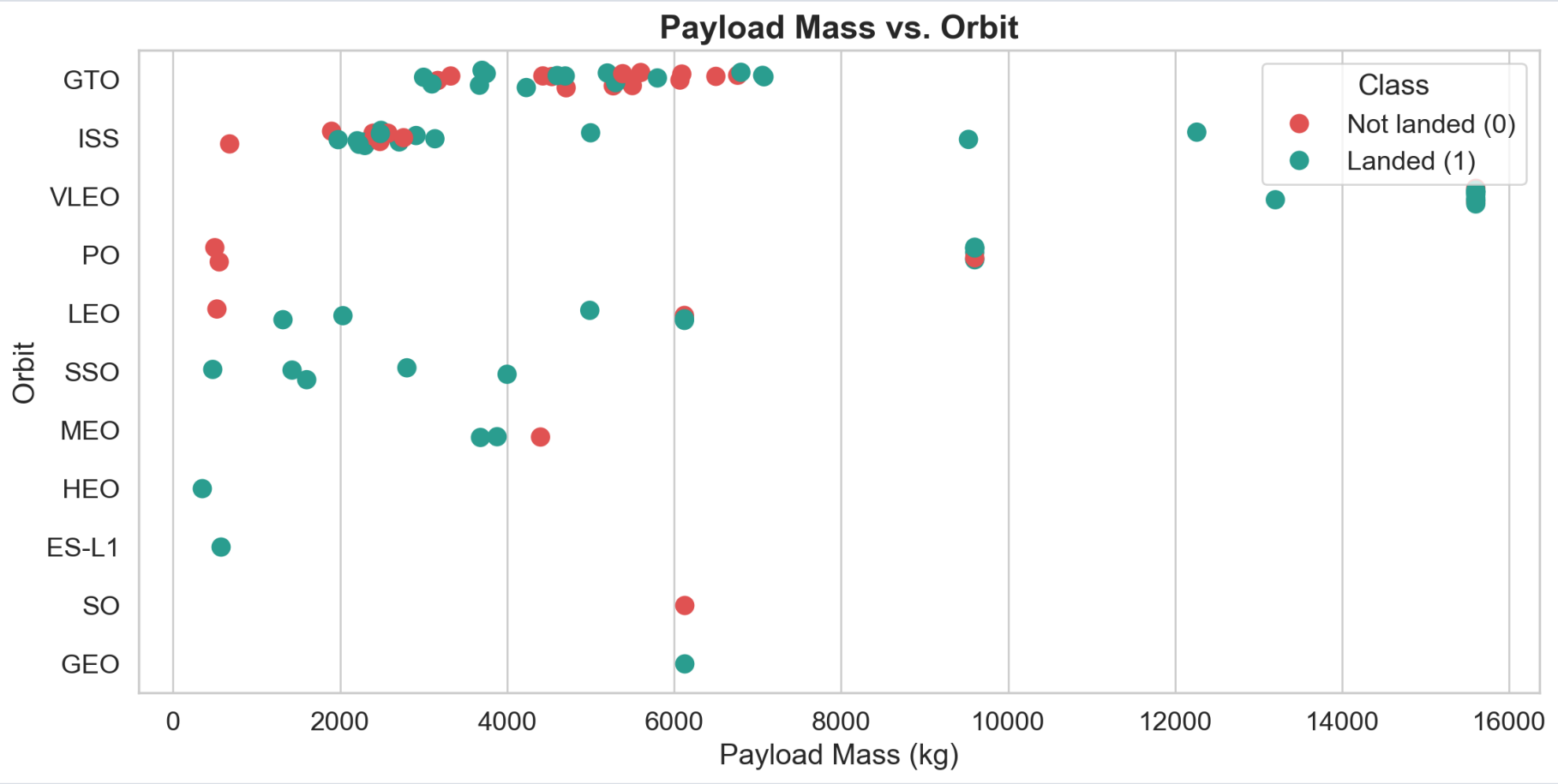
Flight Number vs. Orbit



Insight: LEO failed only in the first two flights, then always landed. GTO shows no clear improvement (52%). VLEO flights all came late and landed 86%.

GitHub: https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/04_eda_visualization.ipynb

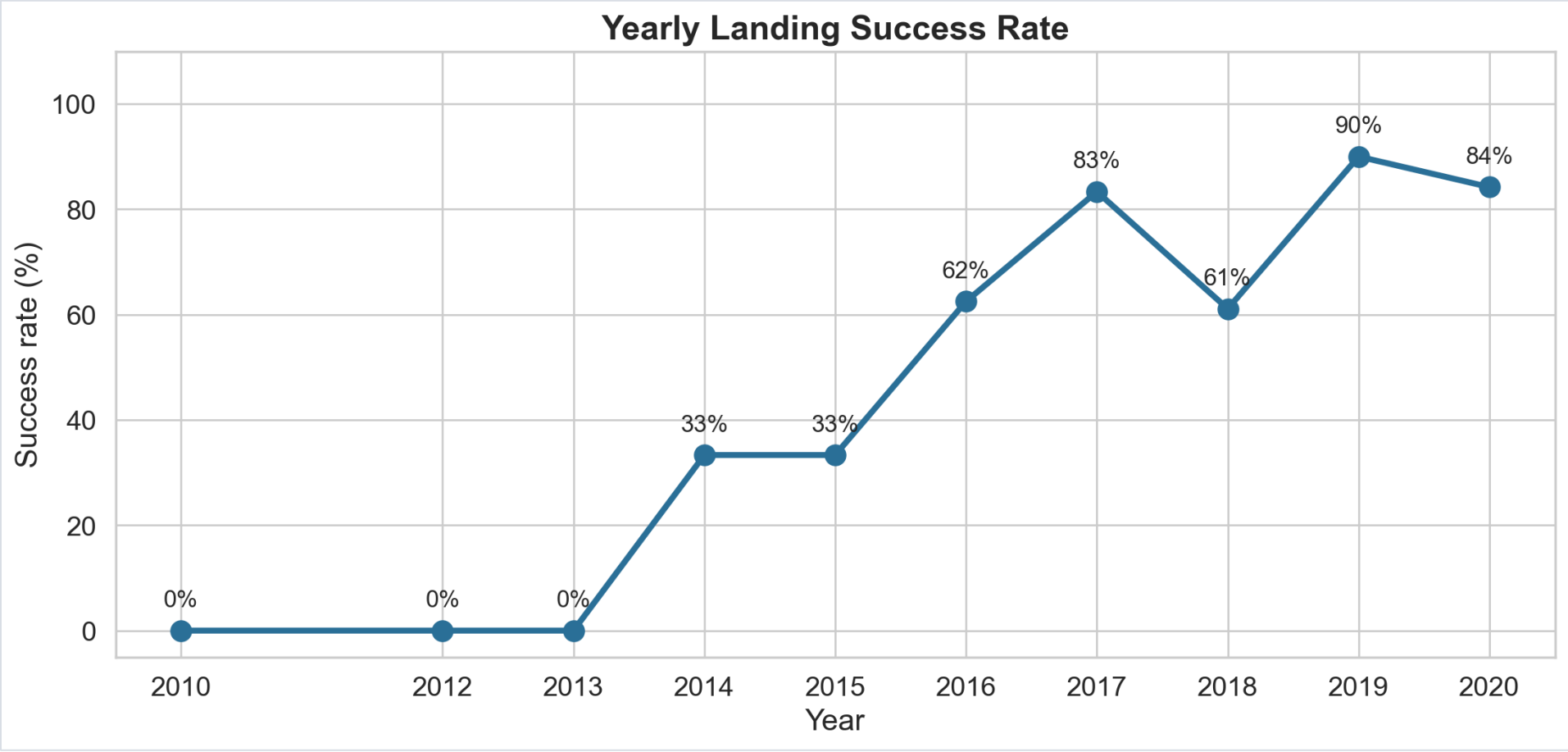
Payload Mass vs. Orbit



Insight: Heavy VLEO payloads (13,200–15,600 kg) land well. In GTO, heavier payloads land less often (62% for ≤ 5,000 kg vs. 43% for > 5,000 kg).

GitHub: https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/04_eda_visualization.ipynb

Launch Success Yearly Trend



Insight: Success rose from 0% (2010–2013) to 90% in 2019 (84% in 2020), with one dip in 2018 (61%).

GitHub: https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/04_eda_visualization.ipynb

SQL Results: Launch Sites

```
SELECT DISTINCT Launch_Site FROM
SPACEXTABLE;
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

```
SELECT * FROM SPACEXTABLE
WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

Date	Booster Version	Launch Site	Payload	PAYLOAD MASS KG	Orbit	Customer	Mission Outcome	Landing Outcome
2010-06-04	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualific	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Insight: 4 unique sites; the first 5 'CCA' launches (2010–2013) were early Dragon / CRS missions with no landing attempt or parachute failures.

GitHub: https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/05_eda_sql.ipynb

SQL Results: Payloads

```
SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTABLE
WHERE Customer = 'NASA (CRS)';
```

Total payload – NASA (CRS)

45,596 kg

```
SELECT AVG(PAYLOAD_MASS__KG_) FROM SPACEXTABLE
WHERE Booster_Version = 'F9 v1.1';
```

Average payload – F9 v1.1

2,928.4 kg

```
SELECT Booster_Version FROM SPACEXTABLE
WHERE PAYLOAD_MASS__KG_ =
  (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE);
```

12 boosters carried the maximum payload of 15,600 kg:

Booster	Booster
F9 B5 B1048.4	F9 B5 B1049.5
F9 B5 B1049.4	F9 B5 B1060.2
F9 B5 B1051.3	F9 B5 B1058.3
F9 B5 B1056.4	F9 B5 B1051.6
F9 B5 B1048.5	F9 B5 B1060.3
F9 B5 B1051.4	F9 B5 B1049.7

GitHub: https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/05_eda_sql.ipynb

SQL Results: Success Rates

```
SELECT Mission_Outcome, COUNT(*) FROM SPACEXTABLE
GROUP BY Mission_Outcome;
```

Mission_Outcome	Total
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Grouped: Success = 100, Failure = 1 → 99.0% mission success

```
SELECT Booster_Version FROM SPACEXTABLE
WHERE Landing_Outcome = 'Success (drone ship)'
  AND PAYLOAD_MASS__KG_ > 4000 AND PAYLOAD_MASS__KG_ <
6000;
```

Booster Version	Payload kg
F9 FT B1022	4696
F9 FT B1026	4600
F9 FT B1021.2	5300
F9 FT B1031.2	5200

Insight: Mission success is very high (100 of 101), but booster landing is the hard part. Only F9 FT boosters landed on a drone ship with 4,000–6,000 kg payloads.

GitHub: https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/05_eda_sql.ipynb

SQL Results: Ranking Landing Outcomes

```
SELECT Landing_Outcome, COUNT(*) AS Total FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome ORDER BY Total DESC;
```

Insight: In the early years, 'No attempt' was most common (10). Drone-ship landings were 50/50 (5 success vs. 5 failure), and ground-pad landings already worked (3 of 3).

Rank	Landing_Outcome	Total
1	No attempt	10
2	Success (drone ship)	5
3	Failure (drone ship)	5
4	Success (ground pad)	3
5	Controlled (ocean)	3
6	Uncontrolled (ocean)	2
7	Failure (parachute)	2
8	Precluded (drone ship)	1

GitHub: https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/05_eda_sql.ipynb

SQL Results: Time Analysis

```
SELECT MIN(Date) FROM SPACEXTABLE
WHERE Landing_Outcome = 'Success (ground pad)';
```

First successful ground-pad landing

22 December 2015

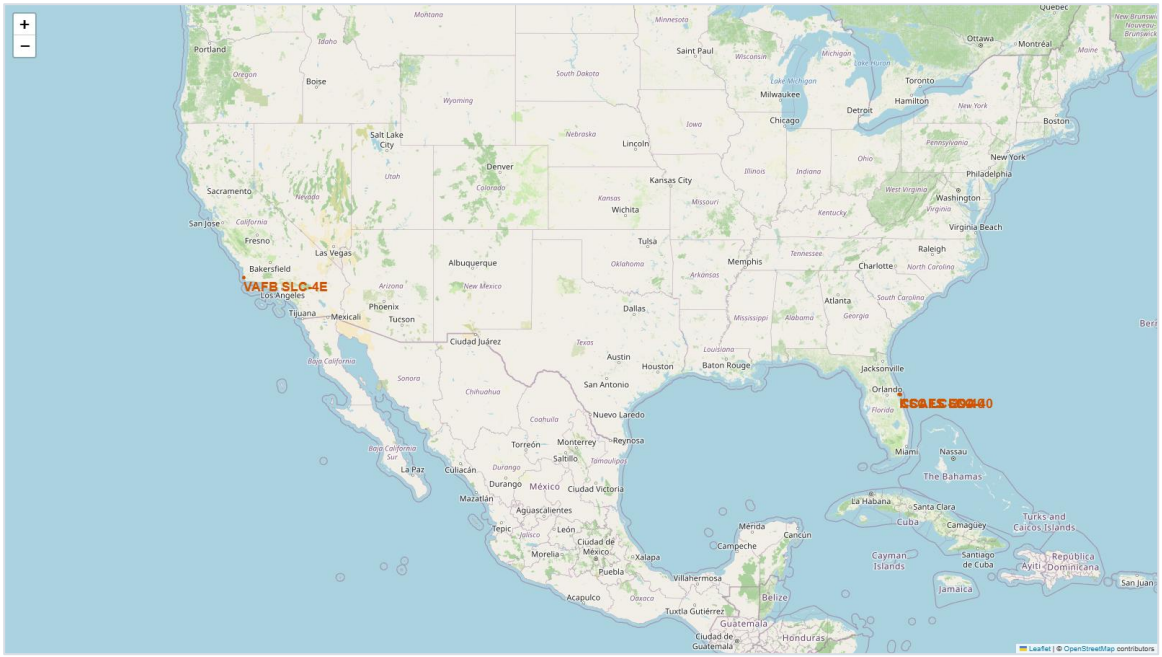
```
SELECT substr(Date,6,2) AS Month, Landing_Outcome,
       Booster_Version, Launch_Site FROM SPACEXTABLE
WHERE Landing_Outcome = 'Failure (drone ship)'
  AND substr(Date,1,4) = '2015';
```

Month	Month name	Landing Outcome	Booster Version	Launch Site
01	January	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	April	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

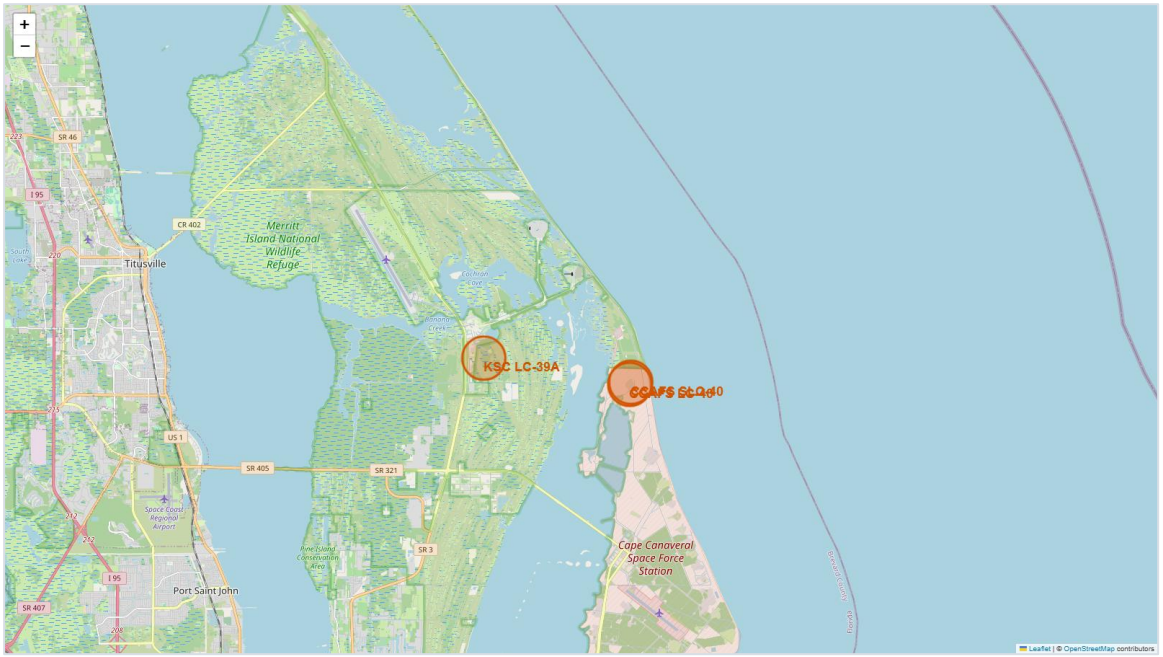
Insight: Both 2015 drone-ship failures were F9 v1.1 boosters from CCAFS LC-40 (January and April). Eight months later (Dec 2015) the first ground-pad landing succeeded.

GitHub: https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/05_eda_sql.ipynb

Folium: All Launch Sites on the Map



All sites (USA)

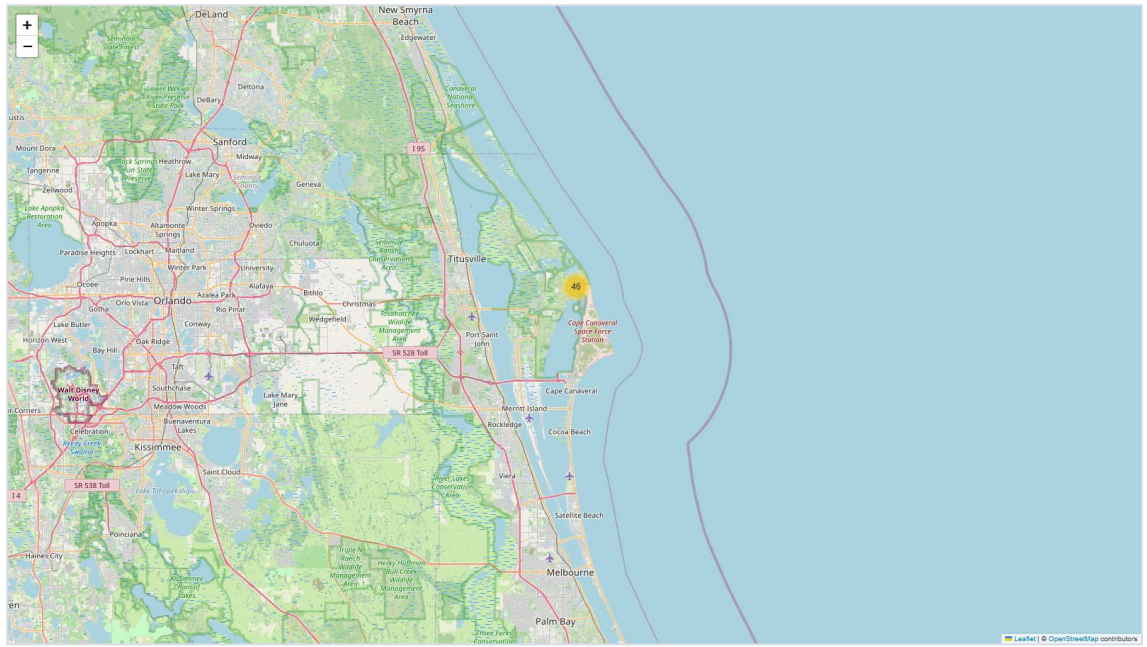


Cape Canaveral close-up (LC-40 and SLC-40 labels overlap: same pad area)

Insight: 3 sites in Florida (CCAFS LC-40, CCAFS SLC-40, KSC LC-39A) and 1 in California (VAFB SLC-4E). All sit on the coast in the south of the USA – closer to the equator gives extra speed from Earth's rotation, and launches fly over the ocean.

GitHub: https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/06_interactive_folium_map.ipynb

Folium: Launch Outcomes per Site



MarkerCluster – number of launches per site

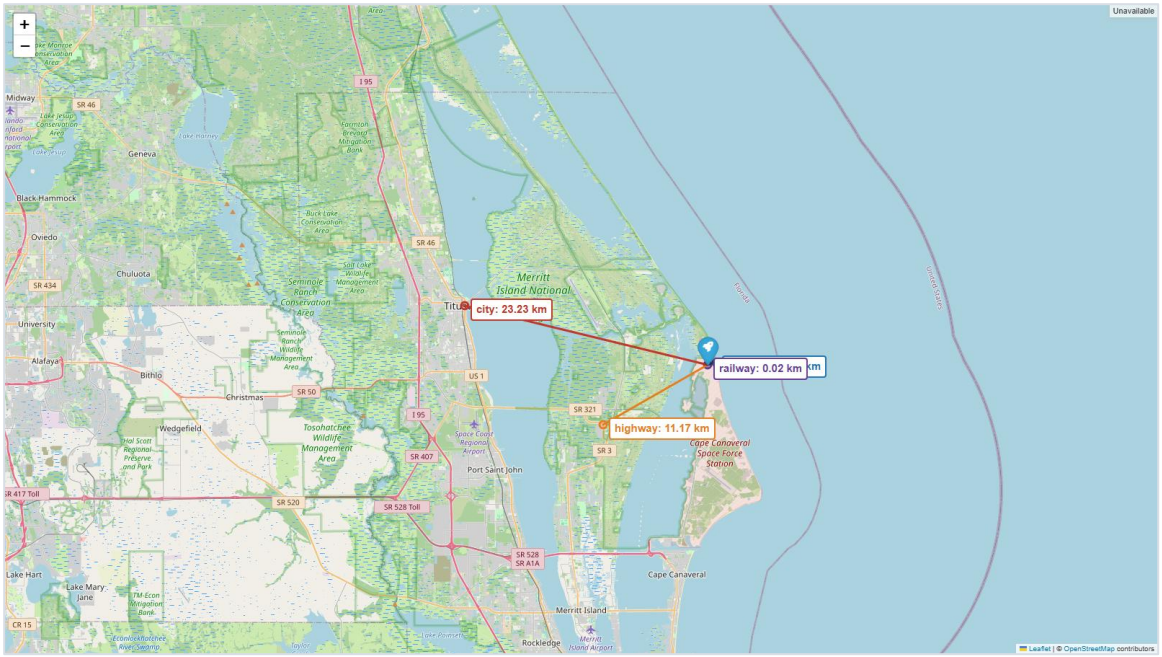


KSC LC-39A cluster opened – green = landed, red = not landed

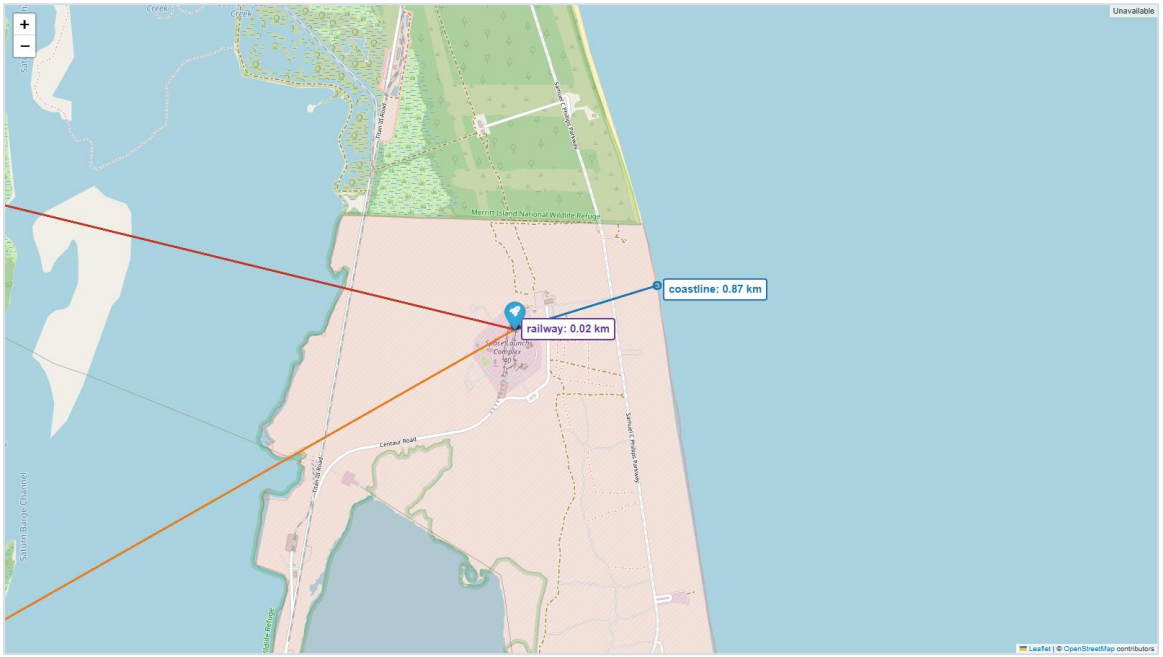
Insight: KSC LC-39A: 10/13 landed (77%) | CCAFS SLC-40: 3/7 landed (43%) | VAFB SLC-4E: 4/10 landed (40%) | CCAFS LC-40: 7/26 landed (27%)

GitHub: https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/06_interactive_folium_map.ipynb

Folium: Proximity Analysis (CCAFS SLC-40)



Overview: highway and city



Close-up: coastline and railway

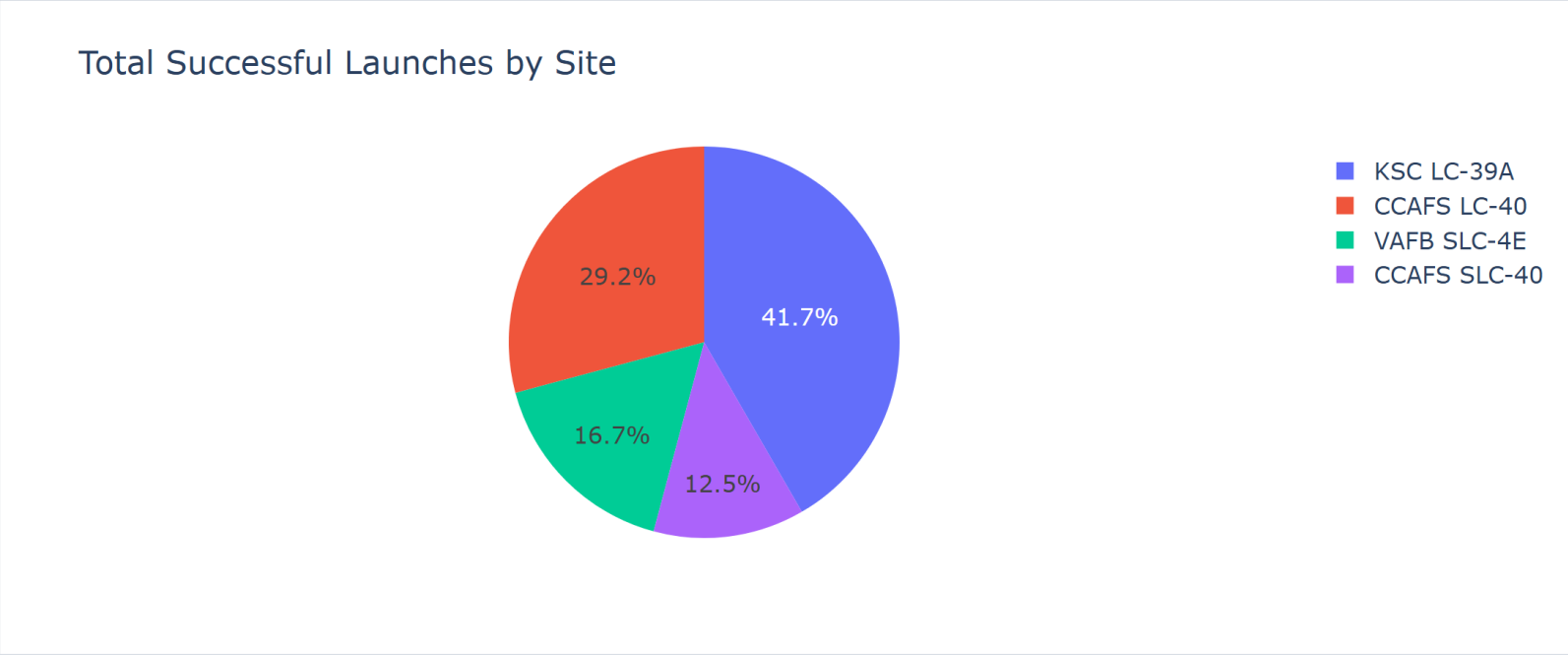
Nearest	Name (OpenStreetMap)	Distance
Coastline	–	0.87 km
Railway	–	0.02 km
Highway	Space Commerce Way	11.17 km
City	Titusville	23.23 km

Findings

- Very close to the coast (0.87 km): a failed rocket falls into the ocean
- Very close to a railway (0.02 km): heavy parts arrive by train
- Highway 11.2 km away: road access for people and supplies
- Nearest city (Titusville) 23.2 km away: far from people for safety

GitHub: https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/06_interactive_folium_map.ipynb

Dash: Successful Launches by Site (All Sites)

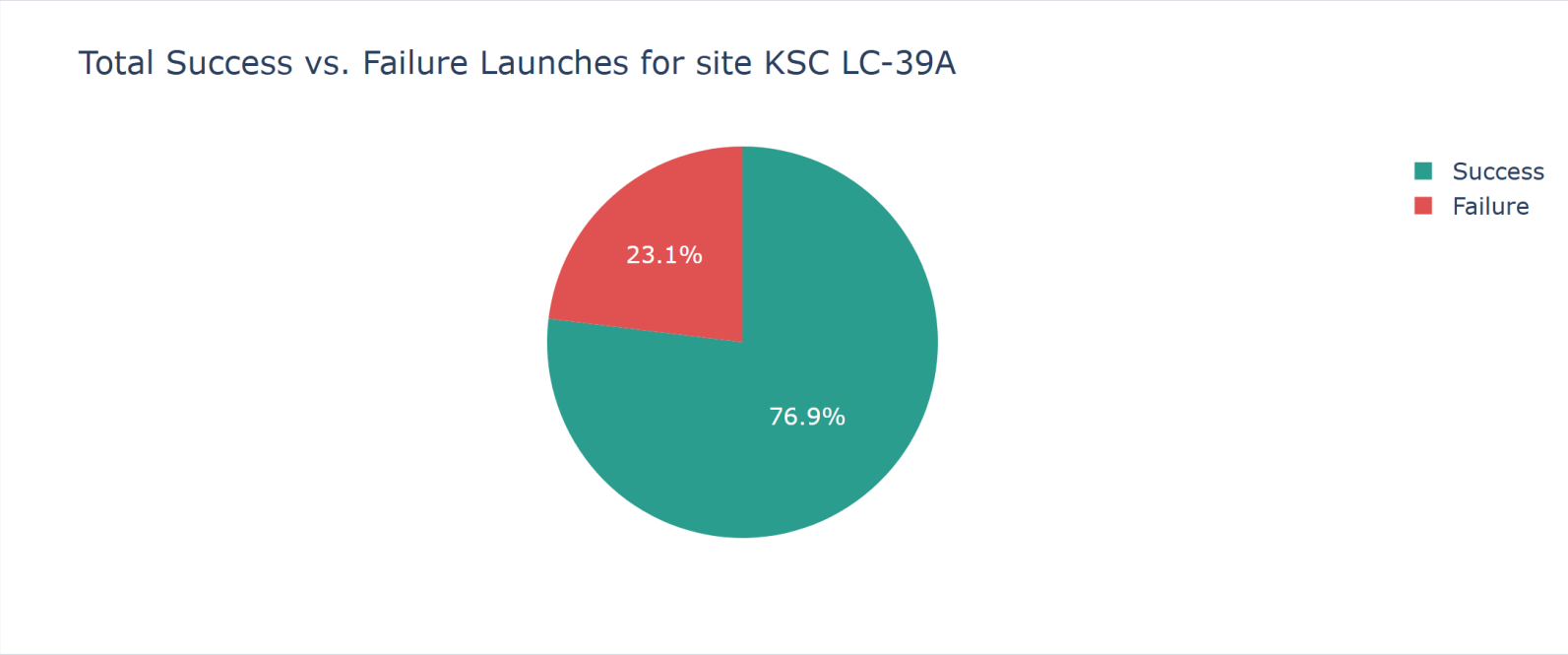


Launch Site	Successes	Share
KSC LC-39A	10	41.7%
CCAFS LC-40	7	29.2%
VAFB SLC-4E	4	16.7%
CCAFS SLC-40	3	12.5%

Insight: KSC LC-39A produced the largest share of all successful launches (41.7%); CCAFS SLC-40 the smallest (12.5%).

GitHub: https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/07_dash_app.py

Dash: Highest Success Rate Site – KSC LC-39A

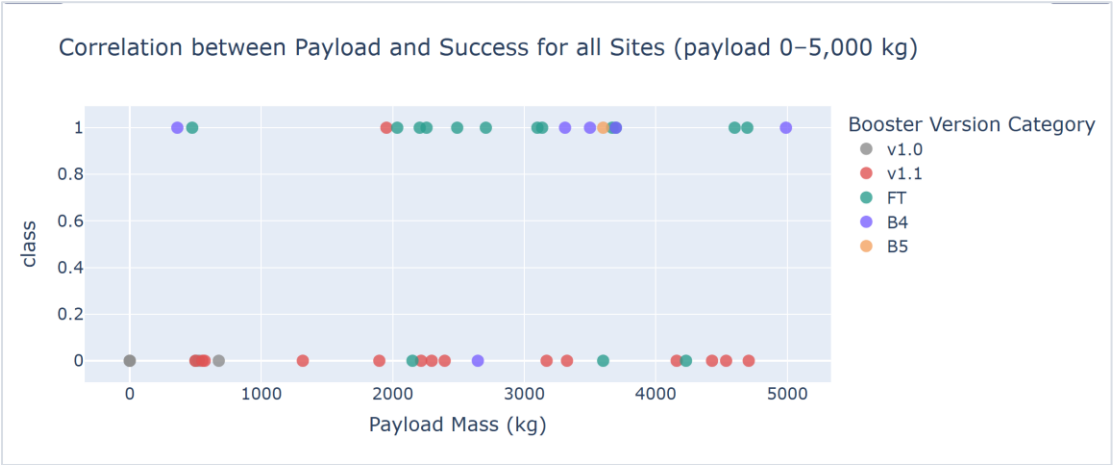


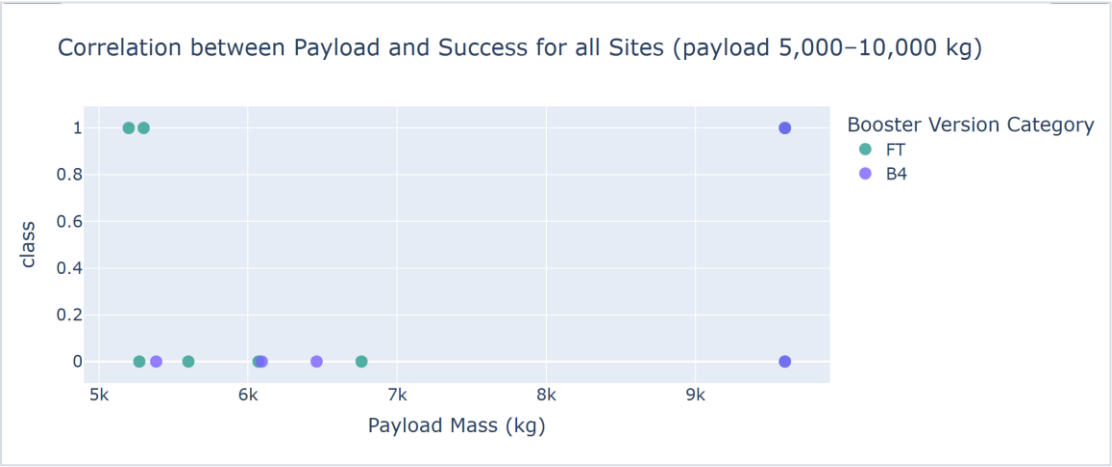
Launch Site	Launches	Success rate
KSC LC-39A	13	76.9%
CCAFS SLC-40	7	42.9%
VAFB SLC-4E	10	40.0%
CCAFS LC-40	26	26.9%

Insight: KSC LC-39A has the highest success rate: 10 of 13 launches (76.9% success, 23.1% failure).

GitHub: https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/07_dash_app.py

Dash: Payload vs. Launch Outcome (All Sites)





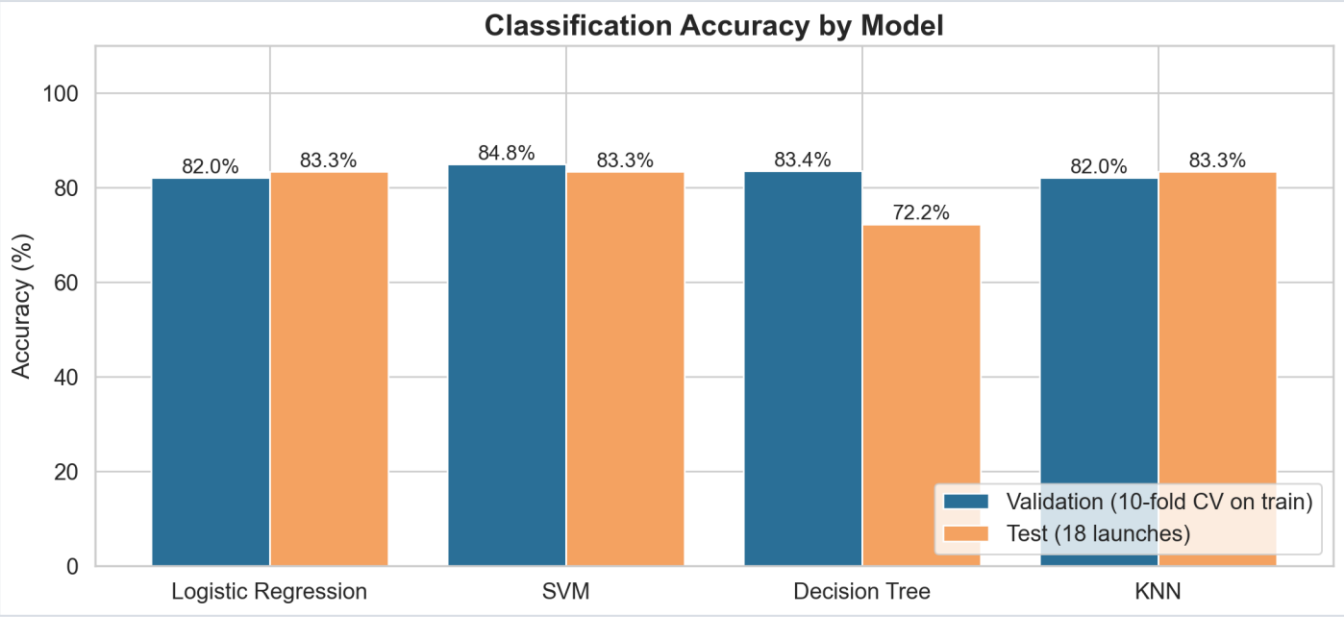
Payload (kg)	Launches	Success
0–2,000	13	23%
2,000–4,000	21	62%
4,000–6,000	13	38%
6,000–8,000	4	0%
8,000–10,000	5	60%

Booster	Launches	Success
B4	11	55%
B5	1	100%
FT	24	67%
v1.0	5	0%
v1.1	15	7%

Insight: Best payload range: 2,000–4,000 kg (62%, n = 21); worst: 6,000–8,000 kg (0 of 4). Best booster: FT (67%, n = 24); B5 is 1 of 1 (too few); v1.1 lowest (7%).

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Classification Accuracy

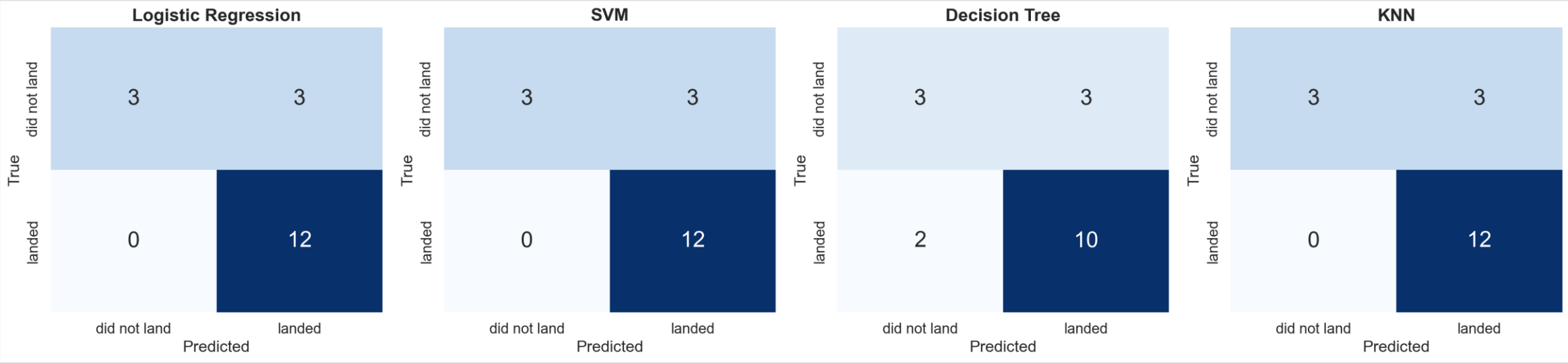


Model	Validation	Test	10-fold CV (all 90)
Logistic Regression	82.0%	83.3%	83.3% ± 12.4
SVM	84.8%	83.3%	84.4% ± 10.2
Decision Tree	83.4%	72.2%	78.9% ± 13.6
KNN	82.0%	83.3%	81.1% ± 8.7

Model	Best parameters (GridSearchCV)
Logistic Regression	C: 0.1, solver: lbfgs
SVM	C: 1, gamma: 0.0316, kernel: sigmoid
Decision Tree	criterion: gini, max_depth: 10, max_features: sqrt, min_samples_leaf: 1, min_samples_split: 2, splitter: random
KNN	algorithm: auto, n_neighbors: 7, p: 1

GitHub: https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/08_predictive_analysis.ipynb

Confusion Matrix



LR, SVM and KNN: same matrix – TP = 12, TN = 3, FP = 3, FN = 0. Every landed booster is found (recall = 100%), but 3 of 6 non-landings are wrongly predicted as landed.

Decision Tree: TP = 10, TN = 3, FP = 3, FN = 2 – it also misses 2 real landings.

> **Insight:** The main error is the false positive: the models are too optimistic about landing. For cost estimates this is the risky mistake (a bid would assume a reusable booster).

GitHub: https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/08_predictive_analysis.ipynb

Best Model

Selected model

Support Vector Machine

Kernel: sigmoid, C = 1, gamma = 0.032

Validation accuracy: 84.8% (best of 4)

Test accuracy: 83.3% (tied best)

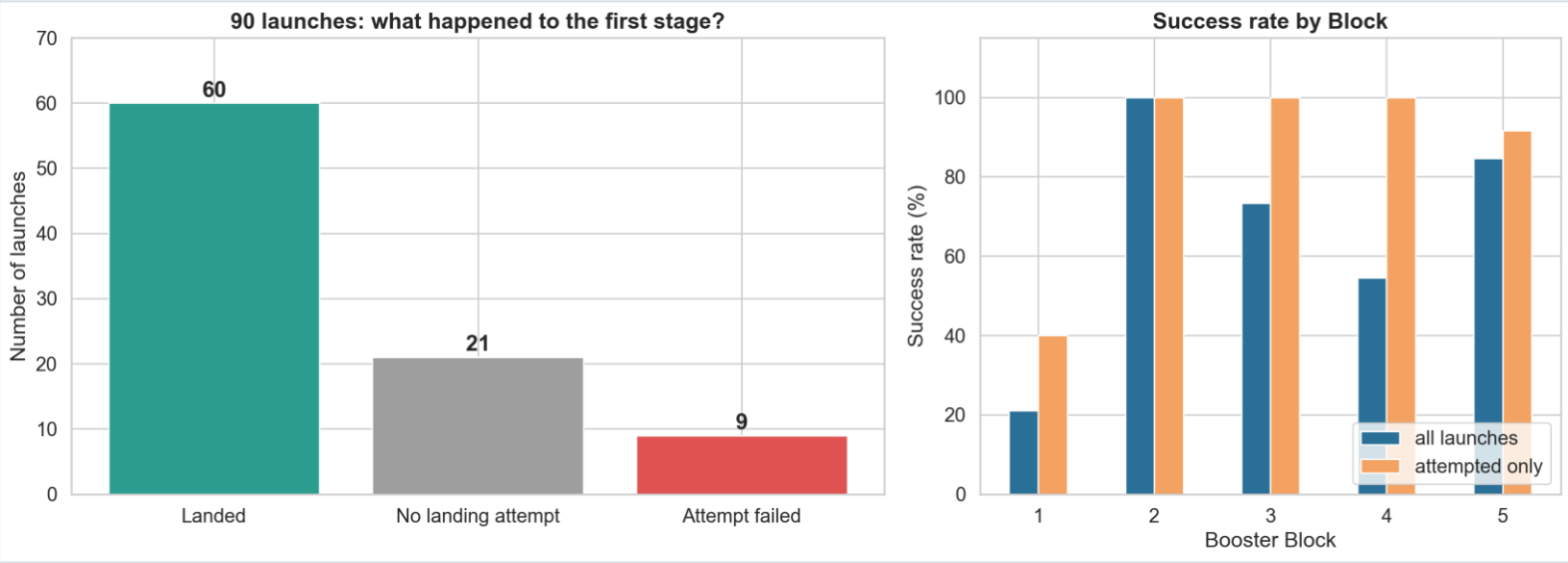
10-fold CV on all data: 84.4% (best of 4)

Why SVM – and an honest warning

- Logistic Regression, SVM, KNN all score 83.3% on the test set: the same 15 of 18 correct.
- With only 18 test launches, one prediction = 5.6 points, so the test set cannot separate these models.
- SVM is highest on both validation (84.8%) and 10-fold CV on all 90 launches (84.4%).
- But the CV spread is about ±10 points, so the gap to Logistic Regression (83.3%) is small.
- Logistic Regression is a good simpler alternative.

GitHub: https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/08_predictive_analysis.ipynb

What Really Drives Landing Success?



Class = 0: 30 launches

- No landing attempt: 21
- Attempt failed: 9

Attempted landings: 60 of 69 succeeded (87.0%)

Attempted, by Block: 1 = 40%, 2–4 = 100%, 5 = 92%

Reused booster: 78% vs. new: 58%

Insight: Most 'failures' are planned: 21 of 30 Class-0 launches had no landing attempt (12 of them GTO). Once SpaceX tries, it lands 87% of the time – so predicting the mission plan (orbit, payload) matters more than the landing itself.

GitHub: https://github.com/khoalee252006/spacex-falcon9-capstone/blob/main/notebooks/08_predictive_analysis.ipynb

Conclusion

- Experience matters: success rose from 0% (2010–2013) to 90% (2019); flights 61–90 land 83% vs. 43% for flights 1–30.
- Orbit matters: VLEO 86%, ISS 62%, GTO only 52% (heavy GTO payloads are hardest).
- Launch sites are all next to the coast, near rail, and far from cities; KSC LC-39A has the best success rate in the dashboard data (77%).
- Payloads of 2,000–4,000 kg and FT boosters show the best success in the dashboard data.
- LR, SVM and KNN reach 83.3% test accuracy; SVM is chosen (best validation / CV), but the small test set means the models are nearly equal.

Creative insight

"Failure" is mostly a choice. 21 of 30 non-landings had no landing attempt; 12 of them were GTO missions.

This also explains the 2018 dip: 6 launches that year did not try to land.

When SpaceX tries, it lands 87% of the time.

Advice for Space Y: price a bid in two steps – (1) will SpaceX try to land for this orbit and payload? (2) if yes, expect the booster to be reused.

GitHub: https://github.com/khoalee252006/spacex-falcon9-capstone

Appendix: Data Sources and Notes

- SpaceX REST API v4 (api.spacexdata.com) – live server returned HTTP 522/525 in Oct 2026; used IBM static copy of /launches/past and archived API responses from web.archive.org (rockets, payloads, launchpads, cores).
- Validation: our dataset_part_1.csv matches IBM's reference in 15 of 17 columns; small differences in PayloadMass and ReusedCount come from a newer API snapshot.
- Wikipedia: List of Falcon 9 and Falcon Heavy launches, revision oldid = 1027686922.
- IBM course files: Spacex.csv (SQL), spacex_launch_geo.csv (Folium), spacex_launch_dash.csv (Dash).
- OpenStreetMap via Overpass API for nearest coastline, railway, highway and city (© OpenStreetMap contributors, ODbL). Note: the course file places KSC LC-39A slightly inland, so we show proximity for CCAFS SLC-40.
- Tools: Python 3.12, pandas, numpy, requests, BeautifulSoup, matplotlib, seaborn, sqlite3, folium, plotly / Dash, scikit-learn, Playwright (screenshots).
- Maps (interactive HTML): images/folium_1_launch_sites.html, folium_2_launch_outcomes.html, folium_3_proximity.html

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